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Review

Treatment of radiation maculopathy and radiation-induced macular edema: A systematic review



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ABSTRACT

Radiation maculopathy and radiation-induced macular edema are common, sight-threatening complications after radiotherapy, especially that used for uveal melanoma. While many treatment and preventive strategies have been proposed, management of these conditions is still challenging. Initially, treatments were based on the use of retinal laser, but the outcomes were poor. Subsequently, management has shifted toward injection of intravitreal antivascular endothelial growth factor or corticosteroids. We reviewed current clinical evidence, which mostly relies on small sample-sized and retrospective studies, for the management of radiation maculopathy and, in particular, radiation-induced macular edema. At present, the first-line approach is usually intravitreal anti-vascular endothelial growth factor. Intravitreal dexamethasone implantation may be an option for those with suboptimal response or contraindications to antivascular endothelial growth factor agents. Possible preventive treatments that require future study are intravitreal bevacizumab and ranibizumab, peripheral laser photocoagulation, and subtenon triamcinolone acetate.

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1. Introduction

Radiation maculopathy represents a common cause of visual loss after radiotherapy for uveal melanoma, especially posterior choroidal melanoma, as well as after radiotherapy of neck and head malignancies.^{24,65} Since the Collaborative Ocular Melanoma Study²⁷ demonstrated similar survival rates between plaque radiotherapy and enucleation for medium

choroidal melanomas, radiation treatments have drawn increasing attention. In particular, plaque brachytherapy and proton beam radiotherapy are considered by many as preferred treatment for small and medium uveal melanomas, while enucleation surgery is reserved for more advanced and larger tumors.³⁴ After this shift toward a globe-sparing approach, the issue of preserving visual function after radiotherapy has become more important.^{11,13} Visual loss from

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